

# Curriculum Vitae

Kishal Tandel

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## Education

- August, 2023 – present **BS-MS in Physical Sciences**, Indian Institute of Science Education and Research, Kolkata, India  
CGPA : 9.18/10 (after 2<sup>nd</sup> Semester)
- 2023 **ISC**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India  
Class 12 - 93.25%, Indian School Certificate (ISC)
- 2021 **ICSE**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India  
Class 10 - 95.40%, Indian Certificate of Secondary Examination (ICSE)

## Research Interests

Quantum optics, Quantum gravity

## Papers

- April 14, 2025 **One-Dimensional Quantum Tunneling**  
Solved one-dimensional time independent Schrödinger equation for all energy regimes, derived the transmission and reflection coefficients and proved their continuity explicitly at  $E = V_0$ , plotted the coefficients using Python and compared the results with classical mechanics qualitatively.

## Projects

- April, 2025 **Quantum Snake**  
Developed a Python-based Quantum Snake model inspired by my work on one-dimensional tunneling, featuring a classic snake, constant potential barriers and a quantum food particle constrained to move in one dimension at a time.
- May, 2025 **Chaotic Trio**  
Implemented the classical fourth-order Runge–Kutta algorithm in C++ to numerically solve Newton's equations for a planar three-body gravitational system and visualized chaotic orbital trajectories under varying initial conditions using custom plotting scripts.

## Skills

- Programming Python, C, C++, Java,  $\text{\LaTeX}$ , Typst, HTML, CSS, JavaScript
- Language English, Hindi, Gujarati